## CAM – Midterm Exam 10/11/2021

### Solutions to Numerical Problems

**METAL FORMING**

A company performs open die forging operations using flat dies on annealed copper samples. The material behaves following a flow curve characterized by $k = 315$ MPa and $n = 0{,}54$ when cold worked, and it has a yield strength equal to 110 MPa when hot worked.

Neglecting the barreling phenomenon, please answer to the following questions:

1. In a press, a cylindrical specimen with initial diameter $d_0 = 25\ \text{mm}$ and initial height $h_0 = 40\ \text{mm}$ is hot worked by applying a compression force $F = 95\ \text{kN}$. As a result, its height is reduced by 30%. Calculate the final diameter of the cylinder and the true strain. Moreover, estimate the friction coefficient between the dies and the specimen.
2. A second press works at room temperature and in the absence of friction thanks to lubrication. In this press, a cylindrical specimen with initial diameter $d_0 = 30\ \text{mm}$ and initial height $h_0 = 50\ \text{mm}$ is compressed to a final height $h_f = 42\ \text{mm}$. Determine the ideal work required by the operation.
3. In the second press (therefore at room temperature and in the absence of friction), a cylindrical specimen with initial diameter $d_0 = 30\ \text{mm}$ and initial height $h_0 = 50\ \text{mm}$ is compressed to fracture. The fracture occurs at a final height $h_f = 28\ \text{mm}$. Calculate the true stress just before the fracture.

***Solution***

1. *In a press, a cylindrical specimen with initial diameter $d_0 = 25\ mm$ and initial height $h_0 = 40\ mm$ is hot worked by applying a compression force $F = 95\ kN$. As a result, its height is reduced by 30%. Calculate the final diameter of the cylinder and the true strain. Moreover, estimate the friction coefficient between the dies and the specimen.*

The final diameter of the cylinder is obtained through the volume conservation relationship:

$$V_0 = A_0 \cdot h_0 = V_f = A_f \cdot h_f$$

$$V_0 = A_0 \cdot h_0 = h_0 \cdot \frac{\pi}{4} d_0^2 = 40 \cdot \frac{\pi}{4} 25^2 = 19635\ \text{mm}^3$$

The cylinder height is reduced by 30%:

$$h_f = h_0 \cdot (1 - 0{,}3) = 40 \cdot 0{,}7 = 28\ \text{mm}$$

Therefore:

$$\varepsilon = \ln {}^{h_f}\!/_{h_i} = \ln {}^{28}\!/_{40} = -0{,}357$$

and

$$A_f = \frac{V_0}{h_f} = \frac{19635}{28} = 701{,}25\ \text{mm}^2$$

$$d_f = \sqrt{\frac{A_f \cdot 4}{\pi}} = \sqrt{\frac{701{,}25 \cdot 4}{\pi}} = 29{,}88\ \text{mm}$$

The friction coefficient between the cylinder and the dies can be derived from the forging force, considering that the average pressure can be evaluated as:

$$p_{av} = Y\left(1 + \frac{\mu \cdot d_f}{3h_f}\right)$$

Considering:

$$F = p_{av} \cdot A_f$$

$$F = Y\left(1 + \frac{\mu \cdot d_f}{3h_f}\right) \cdot A_f$$

It follows:

$$\mu = \frac{(F - Y \cdot A_f) \cdot 3h_f}{Y \cdot A_f \cdot d_f} = \frac{(95000 - 110 \cdot 701{,}25) \cdot 3 \cdot 28}{110 \cdot 701{,}25 \cdot 29{,}88} = 0{,}65$$

2. *A second press works at room temperature and in the absence of friction thanks to lubrication. In this press, a cylindrical specimen with initial diameter $d_0 = 30\,mm$ and initial height $h_0 = 50\,mm$ is compressed to a final height $h_f = 42\,mm$. Determine the ideal work required by the operation.*

The ideal work required by the operation is equal to:

$$W_{ideal} = u \cdot V_0$$

Where:

$$u = \left|\frac{k\varepsilon^{n+1}}{n+1}\right| = \left|\frac{k \ln\left(\frac{h_f}{h_0}\right)^{n+1}}{n+1}\right| = \left|\frac{315 \cdot \ln\left(\frac{42}{50}\right)^{0{,}54+1}}{0{,}54+1}\right| = 13{,}9\ \text{MPa} = 0{,}0139\,\frac{\text{J}}{\text{mm}^3}$$

$$V_0 = h_0 \cdot \frac{\pi}{4} d_0^2 = 50 \cdot \frac{\pi}{4} 30^2 = 35343\ \text{mm}^3$$

Therefore:

$$W_{ideal} = u \cdot V_0 = 0{,}0139 \cdot 35343 = 491{,}3\ \text{J}$$

3. *In the second press (therefore at room temperature and in the absence of friction), a cylindrical specimen with initial diameter $d_0 = 30\,mm$ and initial height $h_0 = 50\,mm$ is compressed to fracture. The fracture occurs at a final height $h_f = 28\,mm$. Calculate the true stress just before the fracture.*

The fracture load can be seen as the flow stress when the specimen breaks. Then, it can be calculated as:

$$\sigma_{fracture} = k \cdot {\varepsilon_{fracture}}^{n}$$

Where:

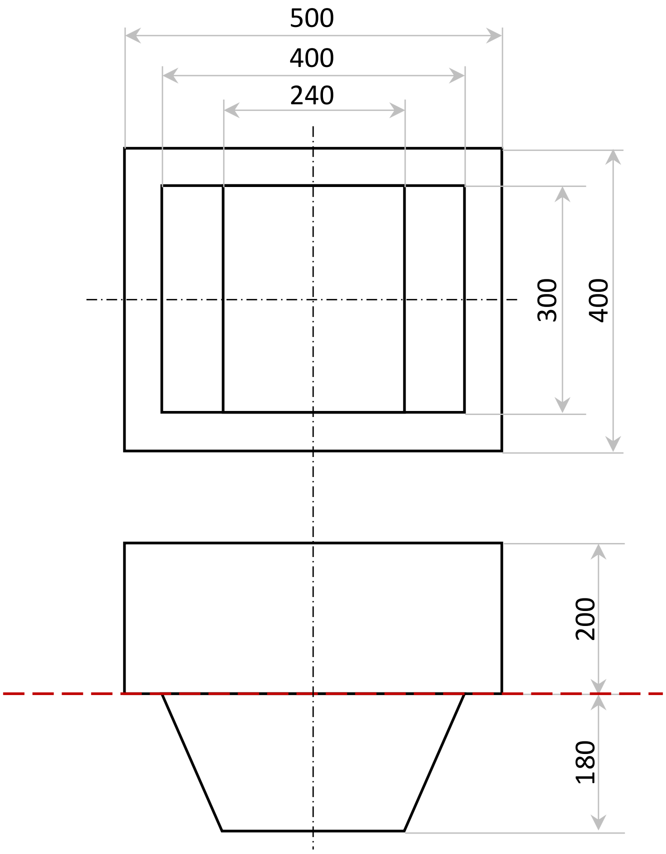
$$\varepsilon_{fracture} = \ln {}^{h_f}\!/_{h_i} = \ln {}^{28}\!/_{50} = -0{,}58$$

Therefore:

$$\sigma_{fracture} = k \cdot {\varepsilon_{fracture}}^{n} = -315 \cdot 0{,}58^{0{,}54} = -234{,}7\ \text{MPa}$$

**METAL CASTING**

In the figure, a pattern to produce a steel part by sand casting is shown (for simplicity, draft angles and fillets/radii are omitted). The cope and drag are identical with the following dimensions: height = 0,5 m, width = 1 m, depth = 1 m.




1. The pattern, and therefore the mould cavity, can be divided into two elementary geometries by the mould parting line (red dashed line). Study the directional solidification of the casting. Is the directional solidification feasible?
2. Knowing that the shrinkage allowance is equal to 1%, calculate the actual total height of the casting at room temperature.
3. Considering: $A_s$:$A_r$:$A_g$ = 4:2:1, c = 0,5, a volume of the casting equal to 58 dm$^3$ (40% in the drag and 60% in the cope), a top riser with a volume equal to 22 dm$^3$, circular gates with diameter equal to 45 mm each. Determine the number of gates of a middle gating system to have a mould filling time not greater than 25 s and the actual mould filling time considering that number of gates.

***Solution***

1. *The pattern, and therefore the mould cavity, can be divided into two elementary geometries by the mould parting line (red dashed line). Study the directional solidification of the casting. Is the directional solidification feasible?*

Let's consider as Body 1 the portion of the cavity inside the cope, and as Body 2 the portion in the drag.

Body 1:

$$V_1 = 400 \cdot 500 \cdot 200 = 40.000.000 \text{ mm}^3$$

$$S_1 = 2 \cdot 400 \cdot 500 - 300 \cdot 400 + 2 \cdot (400 + 500) \cdot 200 = 640.000 \text{ mm}^2$$

$$M_1 = \frac{V_1}{S_1} = \frac{40.000.000}{640.000} = 62{,}5 \text{ mm}$$

Body 2:

$$V_2 = \frac{400 + 240}{2} \cdot 180 \cdot 300 = 17.280.000 \text{ mm}^3$$

$$S_2 = 240 \cdot 300 + 2 \cdot \frac{400 + 240}{2} \cdot 180 + 2 \cdot 300 \cdot \sqrt{\left(\frac{400 - 240}{2}\right)^2 + 180^2} = 305.386 \text{ mm}^2$$

$$M_2 = \frac{V_2}{S_2} = \frac{17.280.000}{305.386} = 56{,}6 \text{ mm}$$

Therefore, solidification starts in Body 2 and end in Body 1.

The ratio $\frac{M_1}{M_2} = \frac{62{,}5}{56{,}6} = 1{,}104$ is the minimum to guarantee a correct solidification.

2. *Knowing that the shrinkage allowance is equal to 1%, calculate the actual total height of the casting at room temperature.*

The actual total height of the casting at room temperature will be

$$H = \frac{200 + 180}{(1 + 0{,}01)} = 376{,}24 \text{ mm}$$

3. *Considering: $A_s$:$A_r$:$A_g$ = 4:2:1, c = 0,5, a volume of the casting equal to 58 dm$^3$ (40% in the drag and 60% in the cope), a top riser with a volume equal to 22 dm$^3$, circular gates with diameter equal to 45 mm each. Determine the number of gates of a middle gating system to have a mould filling time not greater than 25 s and the actual mould filling time considering that number of gates.*

Remembering that:

$$A_{bottleneck} = \frac{V_{total}}{T_{MF} \cdot v}$$

where in this situation $A_{bottleneck} = A_g$

and each single circular gate has an area

$$a_g = \frac{\pi}{4} 45^2 = 1590 \text{ mm}^2$$

The pouring velocity in the bottleneck cross section will be $v = c \cdot \sqrt{2 \cdot g \cdot H_{av}}$, where:

$$H_{av} = \left( \frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2$$

Being $h_i = 500$ mm and $h_f = 0$ mm (we are considering a top riser), we get:

$$r_{drag} = \frac{0{,}4 \cdot V_{casting}}{V_{casting} + V_{riser}} = \frac{0{,}4 \cdot 58}{58 + 22} = 0{,}29$$

$$r_{cope} = \frac{0{,}6 \cdot V_{casting} + V_{riser}}{V_{casting} + V_{riser}} = \frac{0{,}6 \cdot 58 + 22}{58 + 22} = 0{,}71$$

$$H_{av} = \left( \frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2 = \frac{1}{\left( \frac{0{,}29}{\sqrt{500}} + \frac{0{,}71}{\frac{\sqrt{500} + \sqrt{0}}{2}} \right)^2} = 171 \text{ mm}$$

$$v = c \cdot \sqrt{2 \cdot g \cdot H_{av}} = 0{,}5 \cdot \sqrt{2 \cdot 9{,}81 \cdot \frac{171}{1000}} = 0{,}916 \ \frac{\text{m}}{\text{s}} = 916 \ \frac{\text{mm}}{\text{s}} < 1 \frac{\text{m}}{\text{s}}$$

Therefore:

$$A_g = \frac{V_{total}}{T_{MF} \cdot v} = \frac{58.000.000 + 22.000.000}{25 \cdot 916} = 3493 \text{ mm}^2$$

Consequently, it is required a number N of gates equal to:

$$N = \left\lceil \frac{\mathrm{A}_g}{a_g} \right\rceil = \left\lceil \frac{3493}{1590} \right\rceil = \lceil 2{,}2 \rceil = 3$$

The actual mould filling time will be equal to:

$$T_{MF,actual} = \frac{V_{total}}{A_{g,actual} \cdot v} = \frac{58.000.000 + 22.000.000}{3 \cdot 1590 \cdot 916} = 18{,}3 \text{ s}$$